



Scaling Laws for LLMs



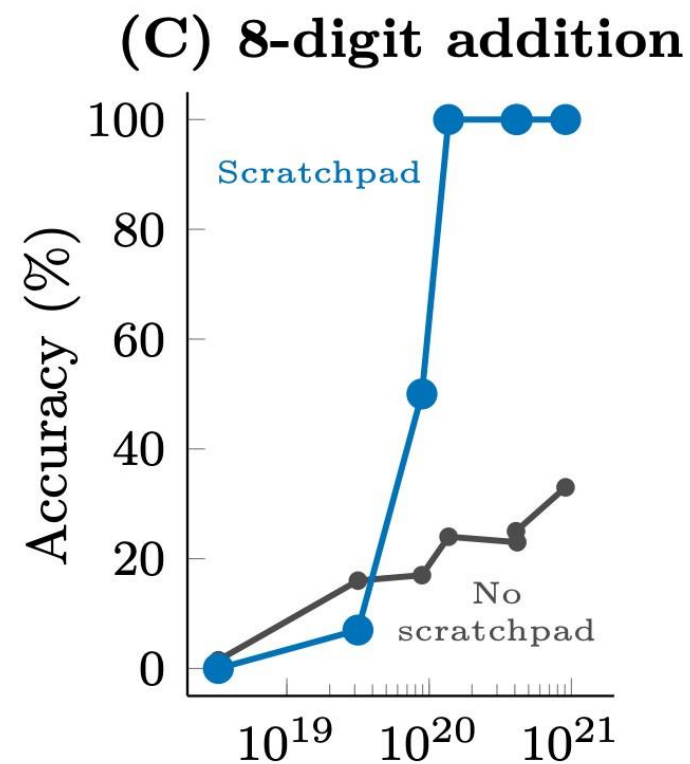
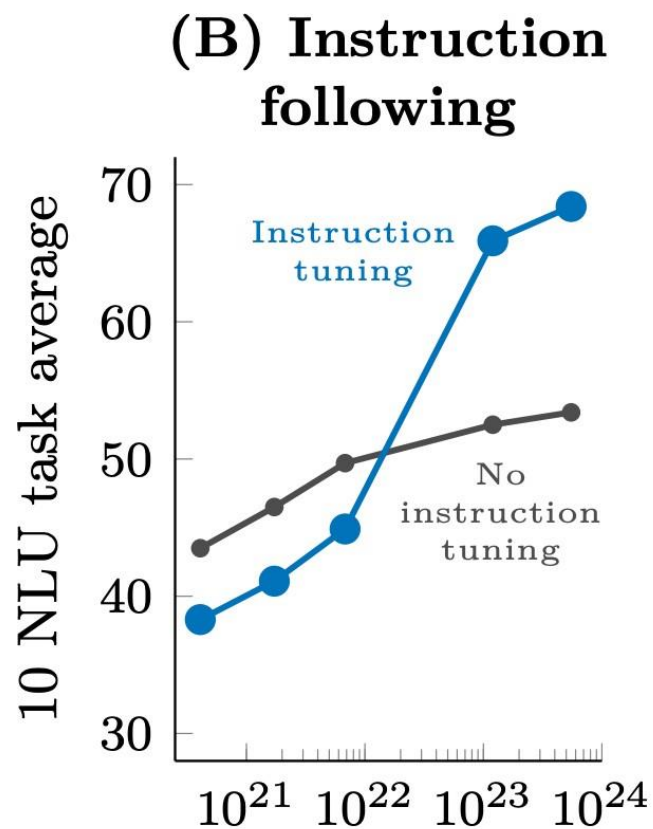
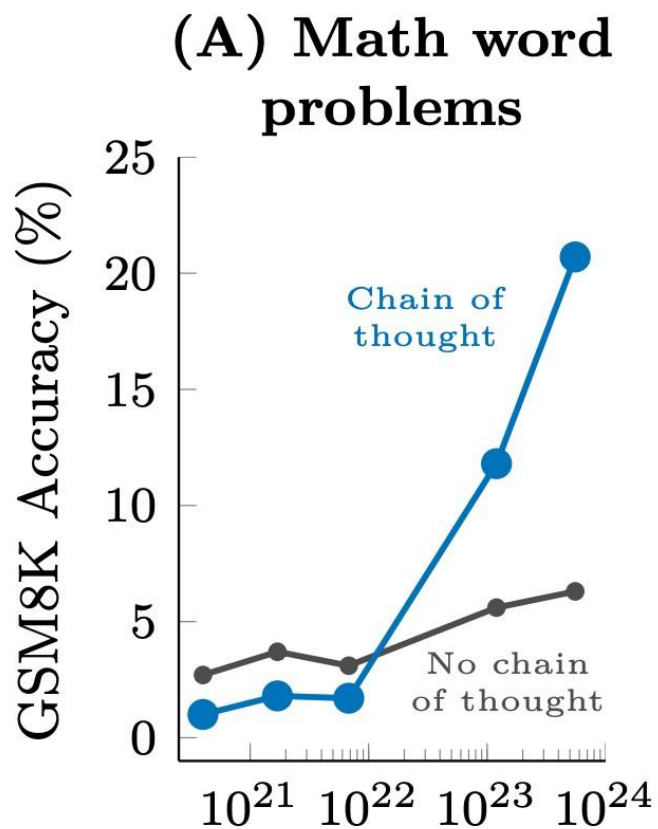
Mohammad Javad Dousti

Sources:

- Mohit Iyyer's Advanced NLP's slides from UMass.

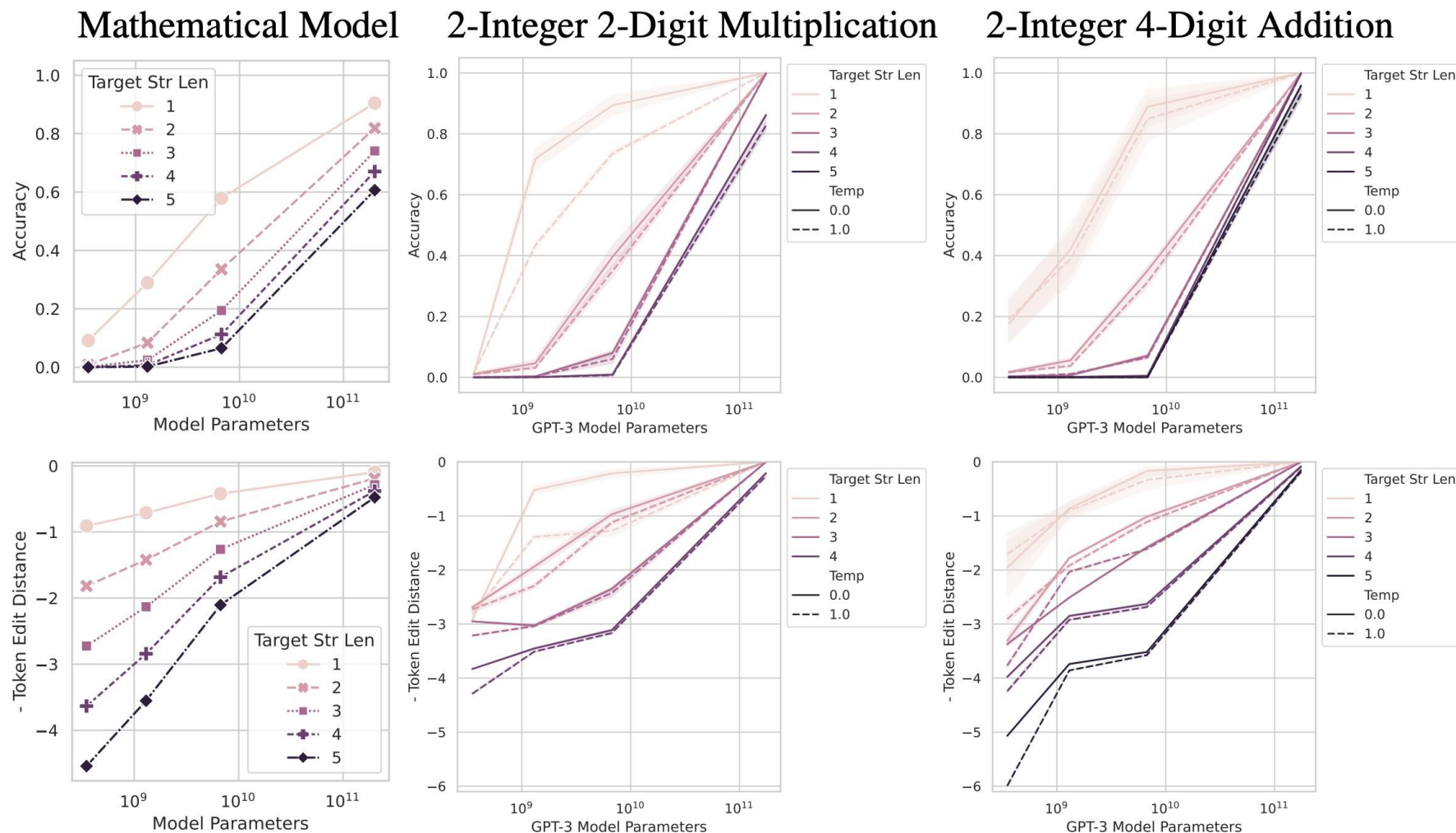
What do bigger LMs buy us?

- ❑ “In-context” learning, chain-of-thought prompting, instruction following, more memorized knowledge and patterns from the training data, etc.
- ❑ Broadly, “**emergent properties**”, which may only appear with larger LMs but not smaller ones
- ❑ *“The ability to perform a task via few-shot prompting is emergent when a model has random performance until a certain scale, after which performance increases to well-above random.” (Wei et al., 2022)*



Model scale (training FLOPs)

Are “emergent properties” really emergent?

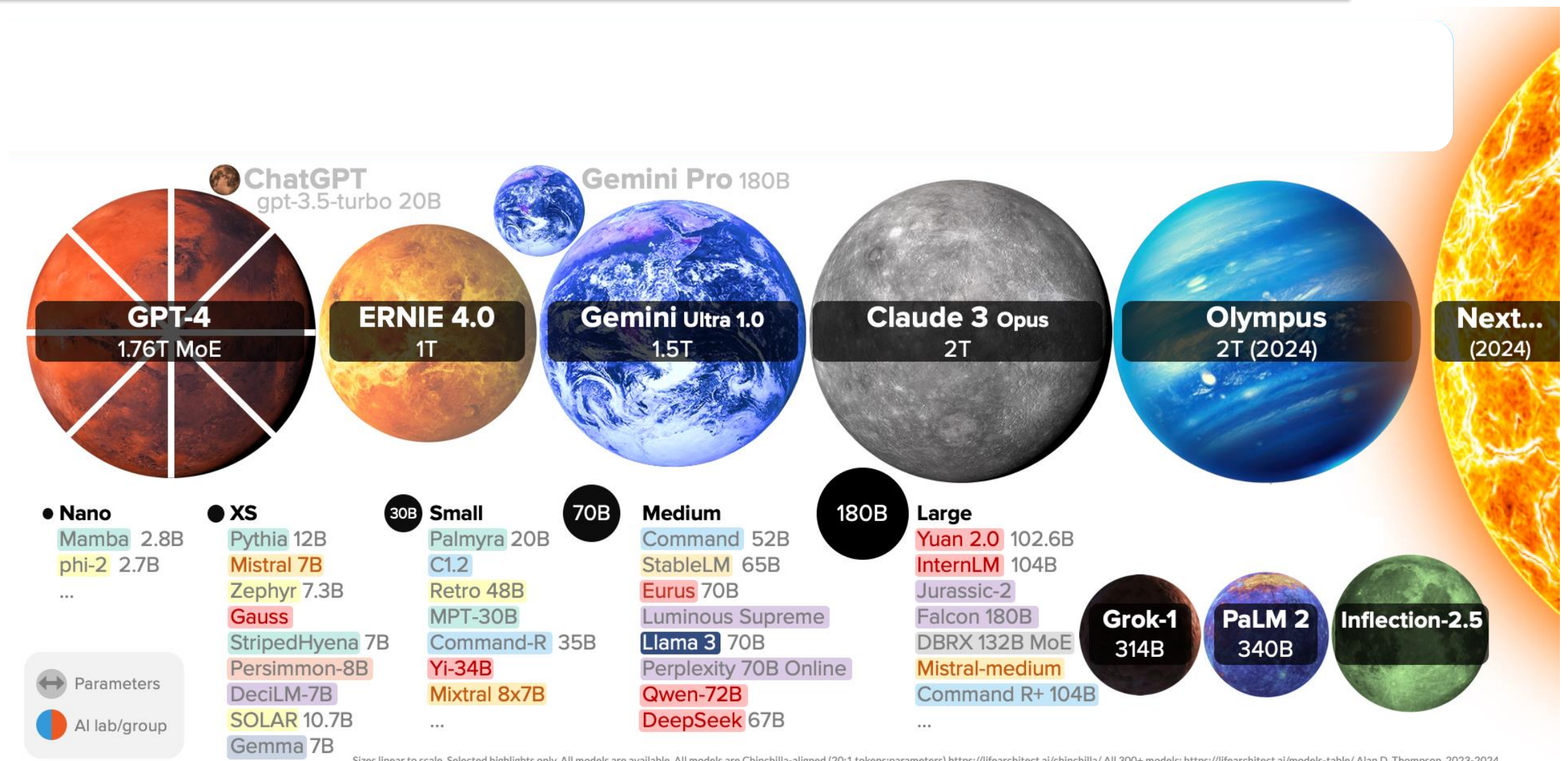


What can we scale?

- ❑ Model size
- ❑ Dataset size
- ❑ Amount of total compute used during training (e.g., number of training steps)



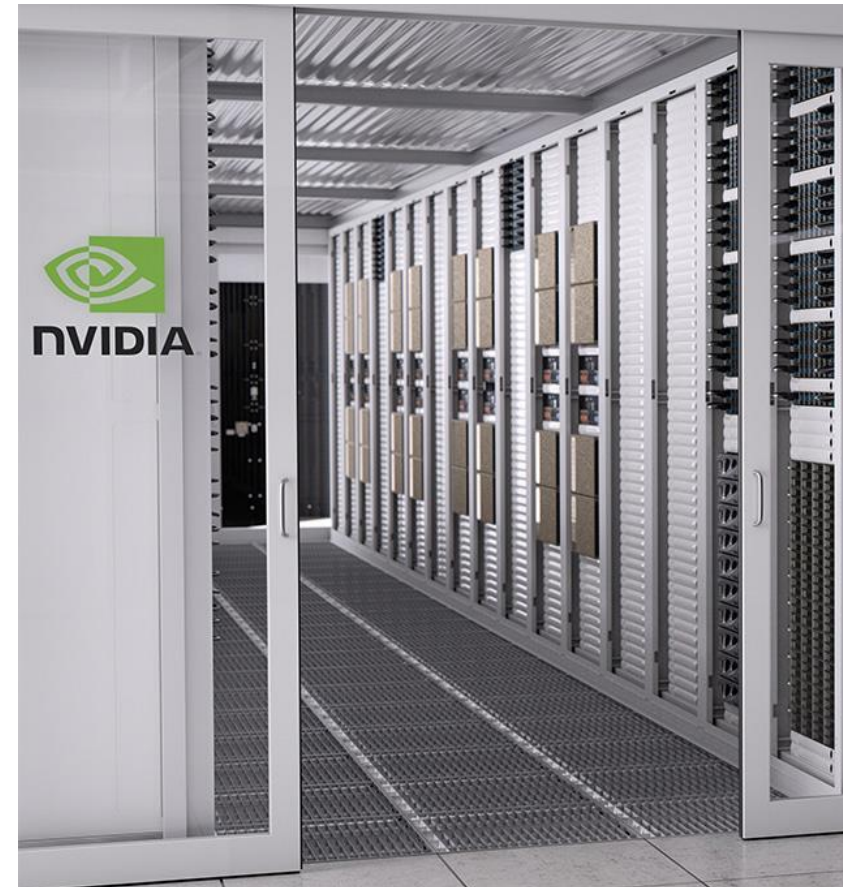
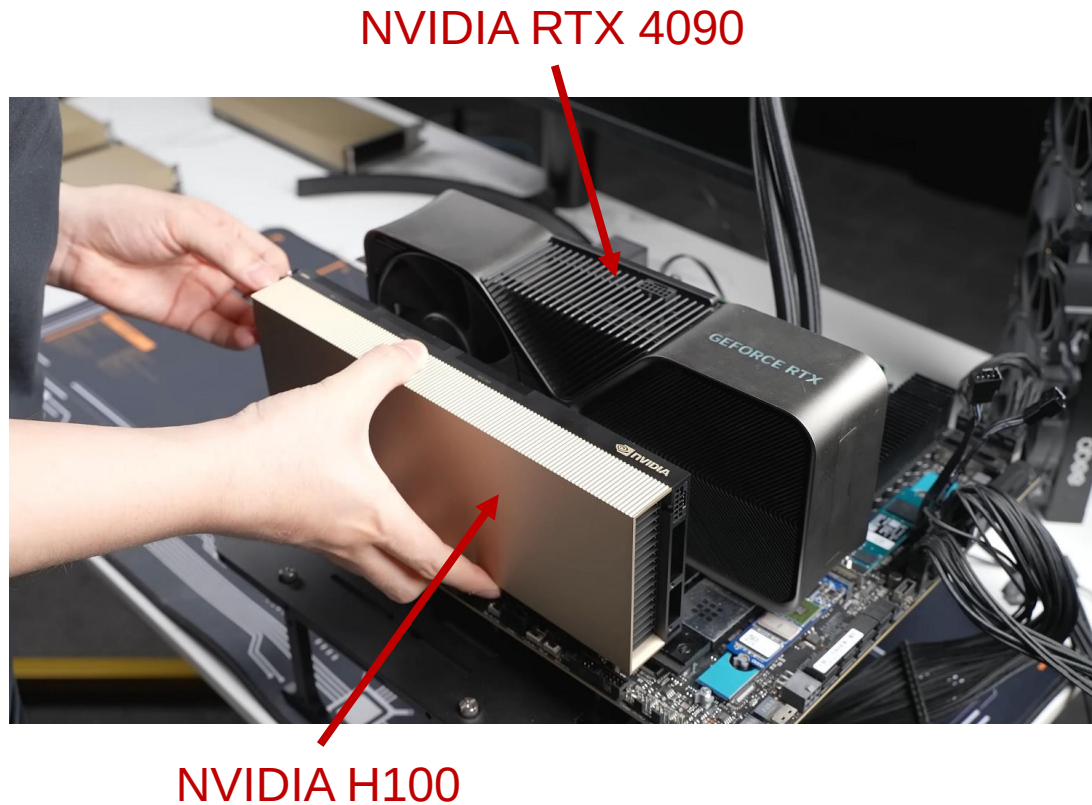
Large Language Model Highlights (Apr/2024)



Sizes linear to scale. Selected highlights only. All models are available. All models are Chinchilla-aligned (20:1 tokens:parameters) <https://lilearchitect.ai/chinchilla/> All 300+ models: <https://lilearchitect.ai/models-table/> Alan D. Thompson. 2023-2024.

Key question

- Given a **fixed compute budget**, what is the optimal model size and training dataset size for training a Transformer LM?



Let's say you can use one GPU for one day

- ❑ Would you train a 5 million parameter LM on 100 books?
- ❑ What about a 500 million parameter LM on one book?
- ❑ Or a 100k parameter LM on 5k books?

Observations from Kaplan et al., 2020

- ❑ Performance depends strongly on scale (model params, data size, and compute used for training), weakly on model shape (e.g., depth, width)
- ❑ Perf vs scale can be modeled with power laws
- ❑ Perf improves most if model size and dataset size are scaled up together. Increasing one while keeping the other fixed leads to diminishing returns
- ❑ Larger models are more sample efficient than smaller models, take fewer steps / data points to reach same loss

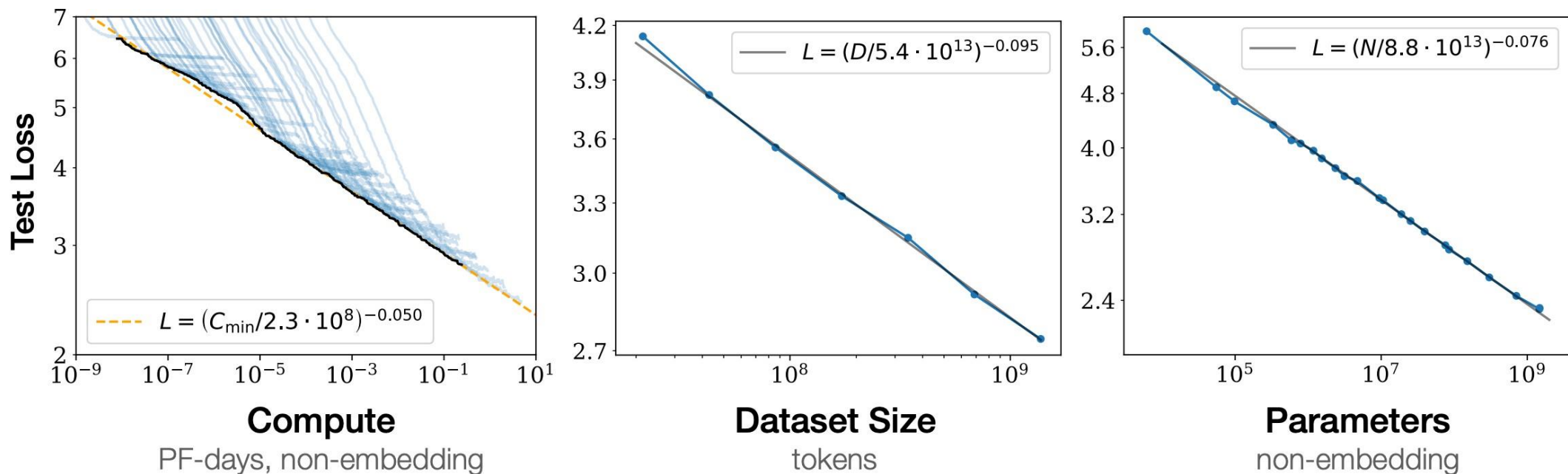
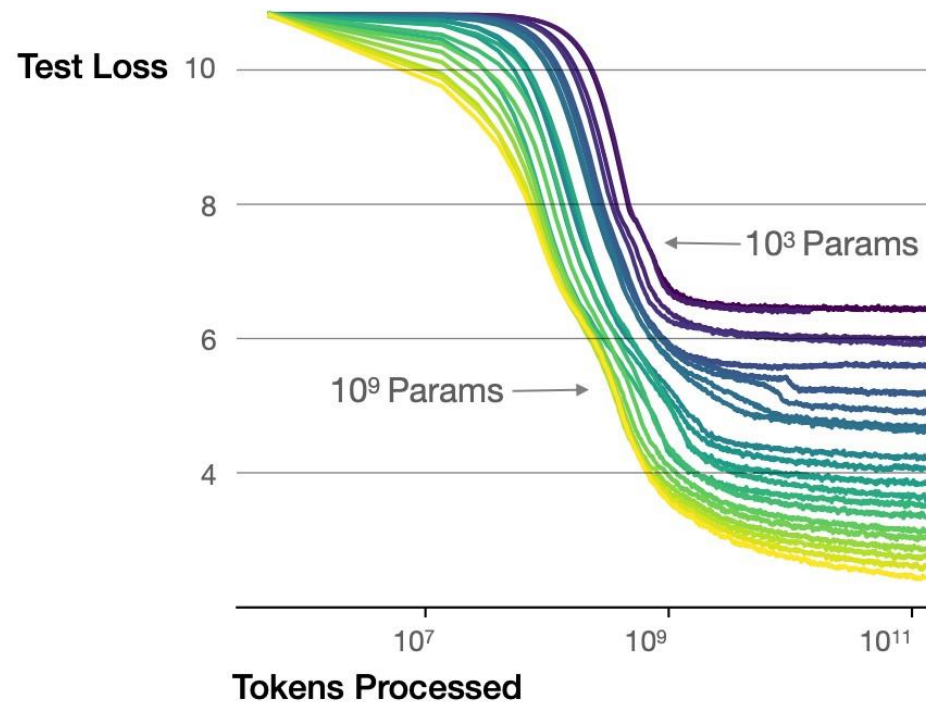


Figure 1 Language modeling performance improves smoothly as we increase the model size, dataset size, and amount of compute² used for training. For optimal performance all three factors must be scaled up in tandem. Empirical performance has a power-law relationship with each individual factor when not bottlenecked by the other two.

Larger models require **fewer samples** to reach the same performance



The optimal model size grows smoothly with the loss target and compute budget

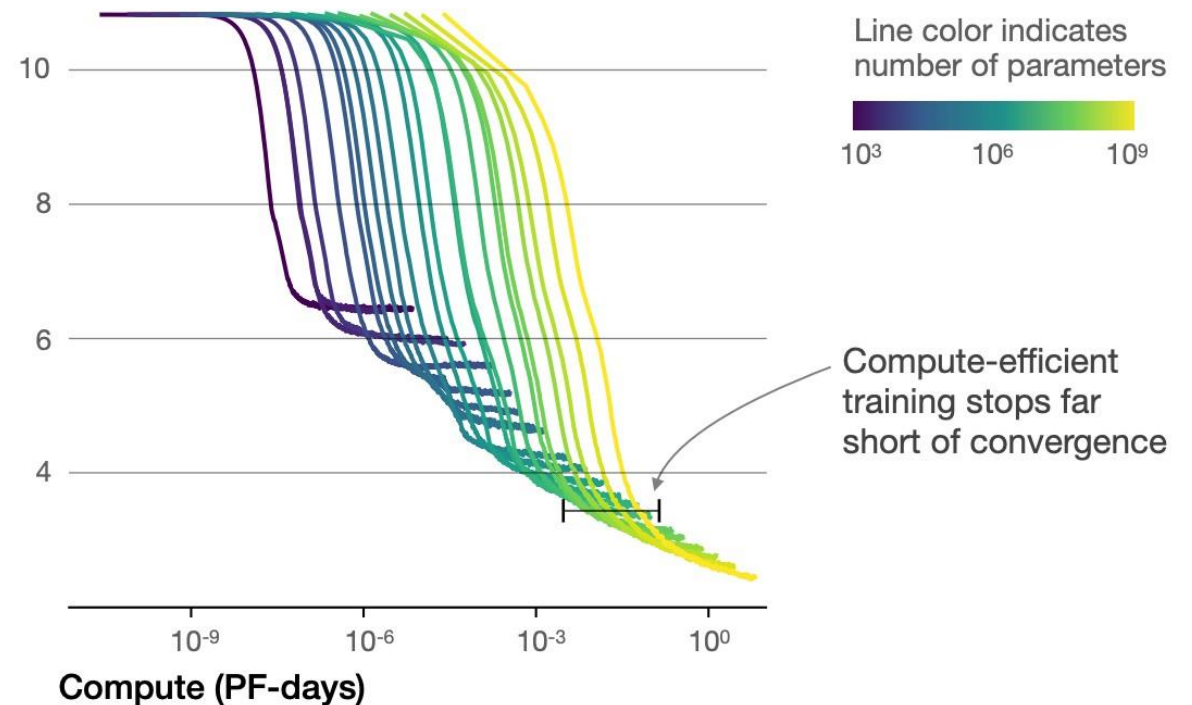


Figure 2 We show a series of language model training runs, with models ranging in size from 10^3 to 10^9 parameters (excluding embeddings).

Issues with Kaplan laws

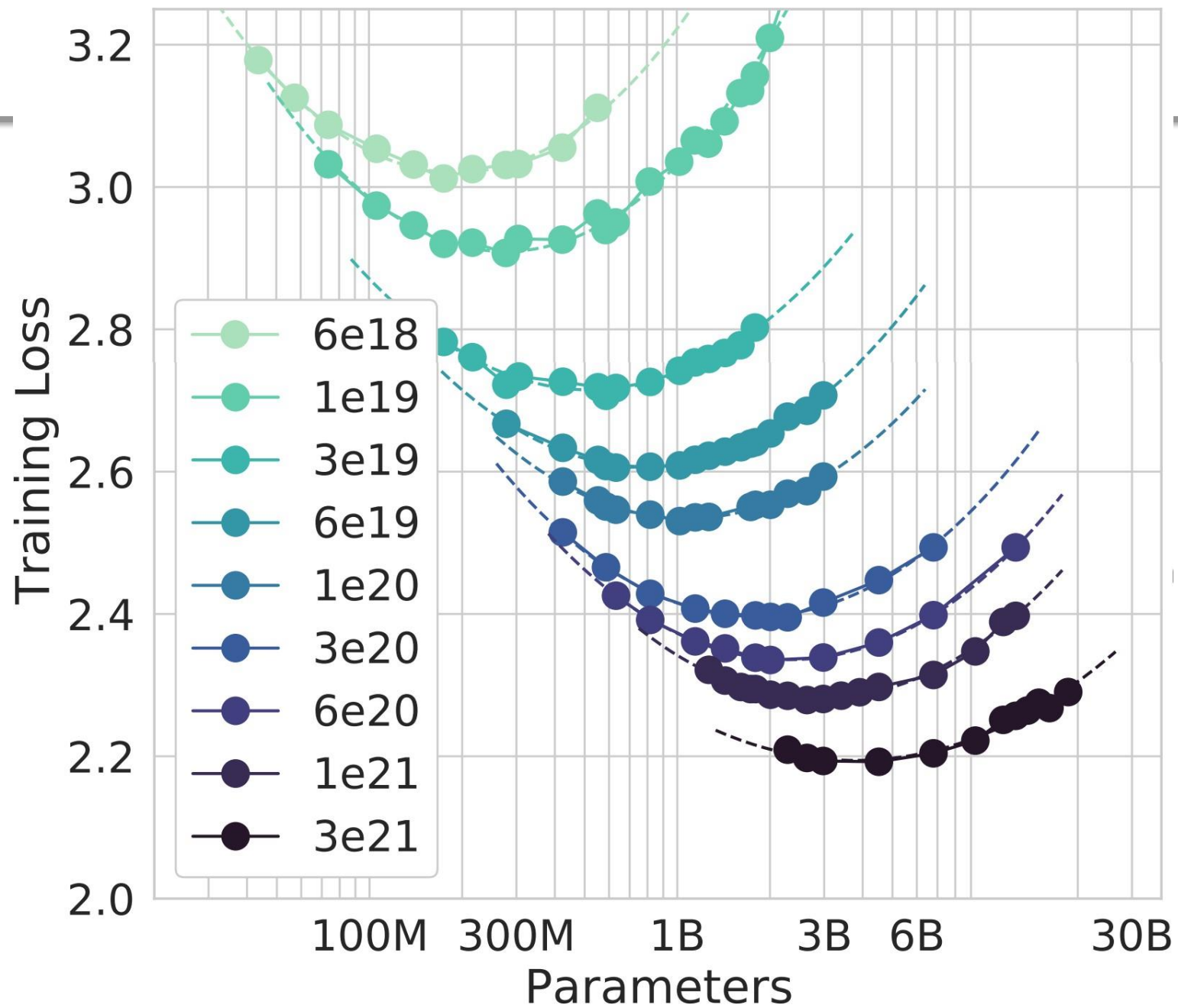
- ❑ Used same learning rate schedule for all training runs, regardless of how many training tokens / batches!
- ❑ This schedule needs to be adjusted based on the number of training steps; otherwise, it can impair performance
- ❑ The resulting “scaling laws” from Kaplan et al., are flawed because of this!

Chinchilla

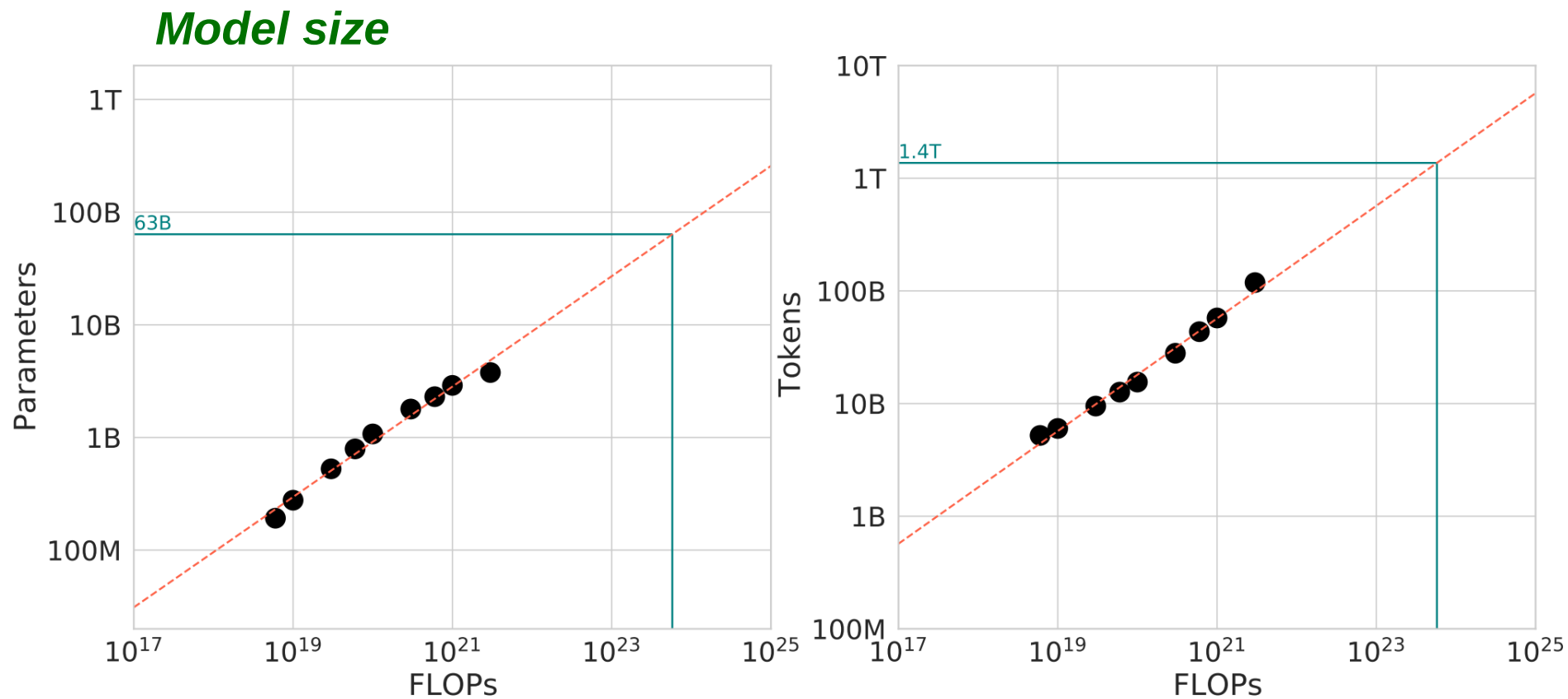
(Hoffmann et al., “Training Compute-Optimal Large Language Models,” 2022)

Quick takeaways

- ❑ Kaplan et al., 2020: if you're able to increase your compute budget, you should prioritize increasing model size over data size
 - With a 10x compute increase, you should increase model size by 5x and data size by 2x
 - With a 100x compute increase, model size 25x and data 4x
- ❑ Hoffmann et al., 2022: you should increase model and data size at the same rate
 - With a 10x compute increase, you should increase both model size and data size by 3.1x
 - With a 100x compute increase, both model and data size 10x



Data size



Chinchilla Scaling Law

- ❑ **L** : LM test loss (average cross-entropy loss)
- ❑ **D** : dataset size (# of training tokens seen)
- ❑ **N** : # of model parameters
- ❑ **C** : computer budget (deterministic function of N and D)
 - $FLOPS(N, D)$
- ❑ Given a fixed compute budget C

Find $\operatorname{argmin}_{N,D} L(N, D)$

Subject to $FLOPS(N, D)=C$

Chinchilla Scaling Law (cont'd)

$$L(N, D) = \frac{A}{N^\alpha} + \frac{B}{D^\beta} + E$$

- E : Test loss achievable by a perfect LM
- $\frac{B}{D^\beta}$: Contribution of the data size
- $\frac{A}{N^\alpha}$: Contribution of the model size
- After training many models, they found the following values:
 - $\alpha = 0.34$, $\beta = 0.28$, $E = 1.69$, $A \cong B \cong 400$
- Experiments:
 - Trained two models with the same compute budget C :
 - **Gopher** (280B params, 300B tokens):
 - $L(\text{Gopher}) = 1.993$
 - **Chinchilla** (70B params, 1.4T tokens):
 - $L(\text{Chinchilla}) = 1.936$



Gopher



Chinchilla

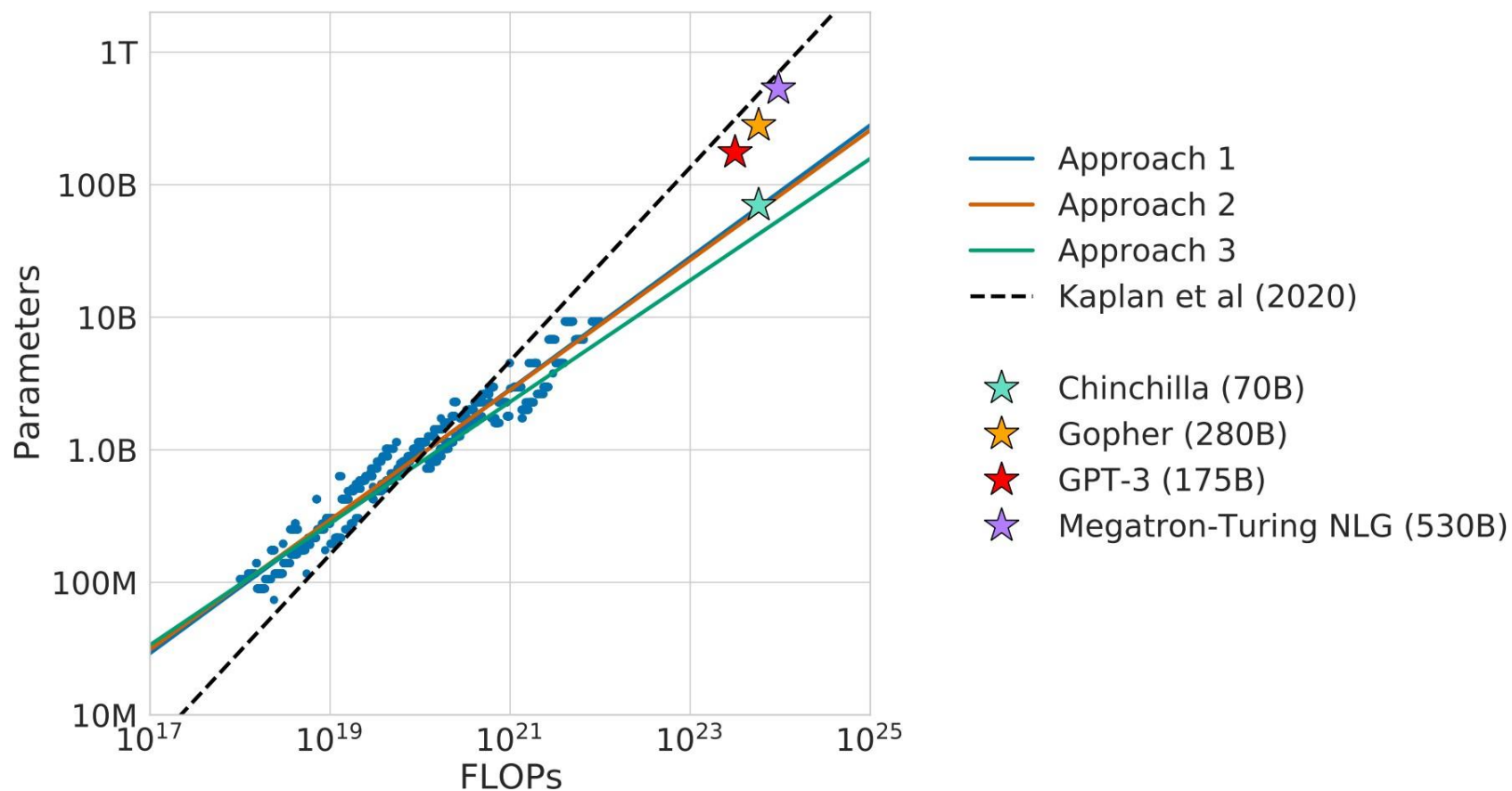
Chinchilla Scaling Law (cont'd)

- ❑ You should increase model and data size at the same rate.
- ❑ Chinchilla scaling law tells us that $D = 20P$
- ❑ It's not recommended to train LLMs for less than 200B tokens though.

-
- Given a fixed compute budget, what is the optimal model size and training dataset size for training a Transformer LM?

Model	Size (# Parameters)	Training Tokens
LaMDA (Thoppilan et al., 2022)	137 Billion	168 Billion
GPT-3 (Brown et al., 2020)	175 Billion	300 Billion
Jurassic (Lieber et al., 2021)	178 Billion	300 Billion
<i>Gopher</i> (Rae et al., 2021)	280 Billion	300 Billion
MT-NLG 530B (Smith et al., 2022)	530 Billion	270 Billion
<i>Chinchilla</i>	70 Billion	1.4 Trillion

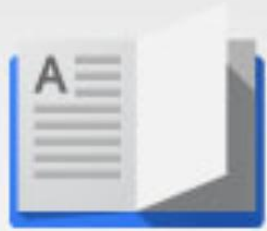
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What about the *type* of data?

What about the type of data?

- ❑ The internet contains a huge amount of text, but it's extremely noisy! Copyrighted text (e.g. published books) are much higher-quality, but is it legal to train on them?
- ❑ What is the impact of repeated data?
 - Repeated data can lead to severe degradation in performance (Brown et al., 2022)
 - “For instance, performance of an 800M parameter model can be degraded to that of a 2x smaller model (400M params) by repeating 0.1% of the data 100 times, despite the other 90% of the training tokens remaining unique.”
 - Repeated data is helpful (Taylor et al., 2022; Galactica)
 - “We train the models for 450 billion tokens, or approximately 4.25 epochs. We find that performance continues to improve on validation set, in-domain and out-of-domain benchmarks with multiple repeats of the corpus.”
 - “We note the implication that the “tokens $\rightarrow \infty$ ” focus of current LLM projects may be overemphasized versus the importance of filtering the corpus for quality.”



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Thursday, August 05, 2010 at 8:26 AM

Posted by Leonid Taycher, software engineer

When you are part of a company that is trying to digitize all the books in the world, the first question you often get is: "Just how many books are out there?"